

Implicit vs. Explicit Brand Attitudes

A Behavioural Analysis of Apple and Samsung

Project Report submitted for the course

Big Data in Behavioural Psychology

by

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Abstract

Consumer attitudes can be examined through both conscious evaluations and automatic associations. While traditional consumer research mainly relies on surveys and ratings, these measures may not fully capture the underlying attitudes that influence how people think about brands. The aim of this study was to compare explicit and implicit attitudes toward two major smartphone brands, Apple and Samsung.

The analysis was based on the Amazon Unlocked Mobile Reviews dataset, which contains more than 400,000 smartphone reviews. After filtering and preprocessing the data, approximately 127,000 reviews related to Apple and Samsung were retained for analysis. Explicit attitudes were measured using customer ratings, whereas implicit attitudes were investigated using Word2Vec word embeddings and the Word Embedding Association Test (WEAT). A discrete-emotion analysis was also conducted to examine the emotional language associated with each brand.

The results showed a difference between explicit and implicit measures. Samsung received slightly higher average ratings, indicating more positive conscious evaluations. In contrast, the WEAT analysis suggested that Apple was associated with more positive concepts within the semantic space of the review corpus. Although the observed WEAT effect was not statistically significant, it pointed toward a moderate tendency for stronger positive implicit associations with Apple. The emotion analysis further revealed that Samsung reviews contained relatively higher levels of sadness and regret-related language.

Overall, the findings support the idea that explicit evaluations and implicit associations may represent different aspects of consumer attitudes. The study also demonstrates how Natural Language Processing and Big Data techniques can be combined with behavioural psychology theories to investigate consumer perceptions using large-scale real-world data.

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CHAPTER 1

Introduction

Consumer attitudes play an important role in shaping how people perceive and evaluate products, brands, and services. In marketing and consumer psychology, these attitudes are often measured through explicit methods such as surveys, questionnaires, and product ratings. Such measures provide valuable information because they capture opinions that consumers consciously express. However, psychological research suggests that people may also hold implicit attitudes that operate automatically and are not always reflected in their self-reported evaluations.

This distinction becomes particularly relevant when examining well-known brands. Consumers may consciously evaluate a smartphone based on practical factors such as quality, price, or performance, while at the same time holding automatic associations related to innovation, prestige, reliability, or personal identity. As a result, explicit evaluations and implicit attitudes may not always point in the same direction.

The growth of online platforms has created new opportunities to investigate consumer attitudes using large-scale behavioural data. Customer reviews contain both explicit evaluations, expressed through ratings, and implicit information embedded within written language. By applying Natural Language Processing (NLP) techniques to these reviews, researchers can explore patterns that may not be visible through traditional survey methods alone.

The present study investigates attitudes toward Apple and Samsung using the Amazon Unlocked Mobile Reviews dataset. Explicit attitudes are examined through customer ratings, while implicit attitudes are explored using Word2Vec embeddings and the Word Embedding Association Test (WEAT). By comparing these two perspectives, the study aims to determine whether consumers display different explicit and implicit attitudes toward the two brands and to interpret these findings within a behavioral psychology framework.

CHAPTER 2

Theoretical Background

2.1 Dual-process Theory

Dual-process theories suggest that human judgment is influenced by two cognitive systems. System 1 operates automatically and quickly through associations and intuition, whereas System 2 involves conscious and deliberate reasoning (Kahneman, 2011). In consumer behaviour, explicit evaluations such as ratings are generally linked to System 2, while implicit attitudes are more closely related to System 1.

2.2 Implicit Attitudes and WEAT

Implicit attitudes are automatic evaluations that may influence behaviour without conscious awareness. Traditionally, psychologists have measured these attitudes using the Implicit Association Test (IAT). In this study, implicit attitudes are examined through the Word Embedding Association Test (WEAT), which applies a similar logic to language data.

2.3 Word Embeddings and Consumer Language

Word embeddings are numerical representations of words based on the contexts in which they appear. Word2Vec is a widely used method that places semantically similar words close together in a vector space (Mikolov et al., 2013). In the context of consumer reviews, word embeddings provide a useful way to investigate implicit brand perceptions.

2.4 Brand-schema Theory

Brand-schema theory proposes that consumers organize knowledge about brands into mental structures containing beliefs, experiences, emotions, and associations. These schemas influence how brands are perceived and evaluated. This perspective helps explain why consumers may hold different implicit associations toward Apple and Samsung even when their explicit evaluations are similar.

CHAPTER 3

Data and Methods

3.1 Dataset

The analysis was conducted using the Amazon Unlocked Mobile Reviews dataset (available on Kaggle), which contains 413,840 customer reviews of smartphone products. Reviews mentioning Apple and Samsung were identified through keyword-based filtering, resulting in 58,197 Apple reviews and 69,923 Samsung reviews (128,120 reviews in total).

During pre-processing, reviews containing insufficient textual information or unusable content were removed. This resulted in a final corpus of 57,610 Apple reviews and 69,258 Samsung reviews, yielding 126,868 reviews for analysis.

The dataset was selected because it provides both numerical ratings and textual feedback, allowing explicit attitudes and implicit language-based associations to be examined within the same source of consumer-generated data. The large sample size makes it particularly suitable for investigating brand perception using Big Data and Natural Language Processing techniques.

Table 3.1: Dataset composition after brand filtering

Brand	Original Reviews	Cleaned Reviews
Apple	58,197	57,610
Samsung	69,923	69,258
Total	128,120	126,868

3.2 Explicit Measure

Explicit attitudes were operationalized using the numerical ratings provided by consumers in the review dataset. Ratings represent conscious evaluations that users deliberately assign to products and therefore serve as a direct measure of explicit attitudes. For each brand, the mean rating was calculated and compared using statistical tests to determine whether significant differences existed between Apple and Samsung reviews. By analyzing rating distributions and average scores, it was possible to assess how consumers consciously evaluated the two brands. This analysis provided the explicit component of

the study, which was later compared with the implicit measures derived from textual data.

3.3 Text Pre-processing and Word2Vec Training

Review texts were converted to lowercase, tokenized, and cleaned by removing punctuation, numbers, and other non-linguistic elements. The cleaned corpus was then used to train a Word2Vec Skip-gram model with 100-dimensional vectors, a context window of 5 words, a minimum word frequency of 20, and 20 training epochs.

The resulting word embeddings represented semantic relationships between words, with similar words positioned closer together in the vector space. These embeddings formed the basis for the implicit attitude analyses.

3.4 WEAT Implementation and Emotion Analysis

Implicit attitudes were measured using the Word Embedding Association Test (WEAT). Apple and Samsung served as target concepts, while positive and negative evaluative words were used as attribute categories. WEAT assessed the relative association of each brand with positive and negative concepts within the embedding space. Effect sizes and permutation tests were used to evaluate the strength of these associations.

To complement the WEAT analysis, a discrete-emotion analysis was conducted using emotion categories including joy, trust, sadness, anger, and regret. The relative frequency of emotion-related terms was compared across brands to identify differences in emotional expression within consumer reviews.

CHAPTER 4

Results

4.1 Explicit Attitudes

Explicit attitudes were assessed through customer ratings. The analysis showed that Samsung received a slightly higher average rating (3.97) than Apple (3.92). Statistical testing indicated that this difference was significant, suggesting that consumers consciously evaluated Samsung somewhat more positively than Apple.

Although the difference in mean ratings was relatively small, the large sample size provided sufficient statistical power to detect reliable differences between the two brands. These findings suggest that, at the level of conscious evaluation, Samsung was perceived more favorably by consumers within the review dataset.

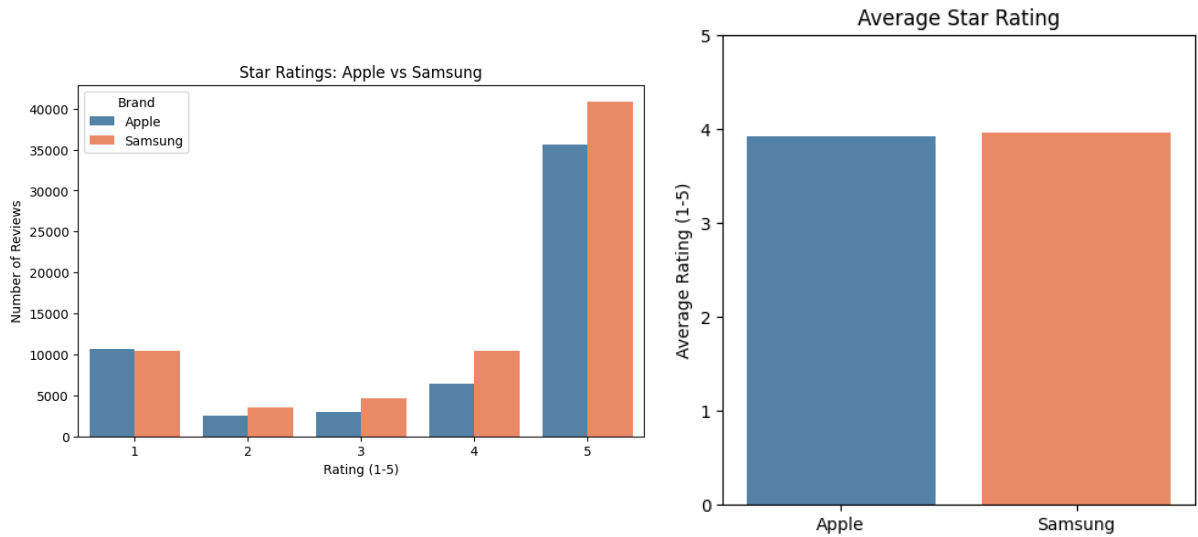


Figure 4.1: Explicit attitudes toward Apple and Samsung. The left panel shows the distribution of customer ratings, while the right panel shows the average rating for each brand. Samsung received slightly higher ratings overall.

4.2 Implicit Attitudes (WEAT)

Implicit attitudes were examined using the Word Embedding Association Test (WEAT). The analysis produced a positive effect size (Cohen's $d = 0.45$), indicating that Apple exhibited stronger associations with positive attribute words than Samsung within the embedding space. This suggests a moderate tendency for Apple to be linked with more favorable implicit associations in consumer language.

Table 4.1: WEAT Results for Apple and Samsung

Metric	Value
Effect Size (Cohen's d)	0.449
p-value	0.587
Number of Permutations	70

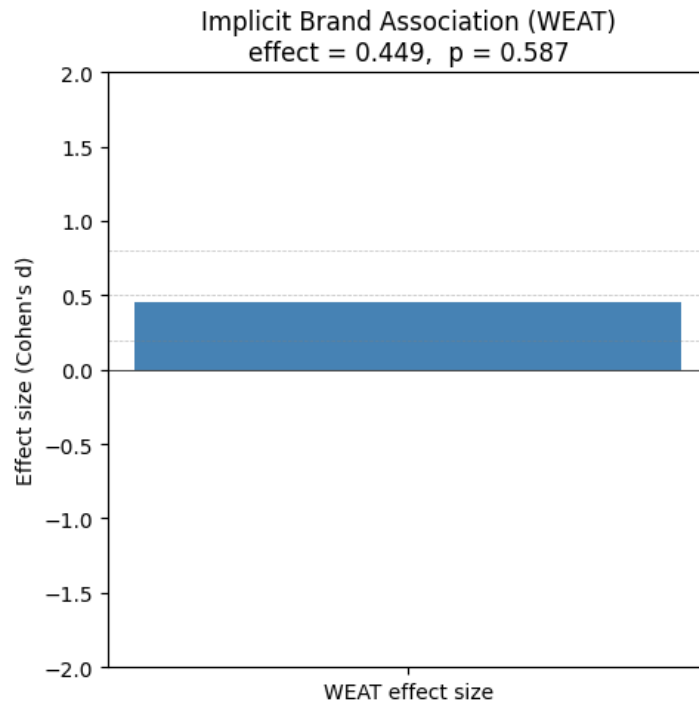


Figure 4.2: WEAT effect size (Cohen's d) for implicit brand associations.

However, the permutation test yielded a non-significant result ($p = 0.587$). This outcome should be interpreted cautiously, as the limited number of possible permutations (70 in total, given 4+4 target words) reduced the statistical power of the test. Consequently, the absence of statistical significance does not necessarily imply the absence of an effect.

Overall, the WEAT results provide preliminary evidence that Apple may possess stronger positive implicit associations than Samsung, despite Samsung receiving slightly higher explicit ratings. This divergence between explicit and implicit measures represents one of the central findings of the study.

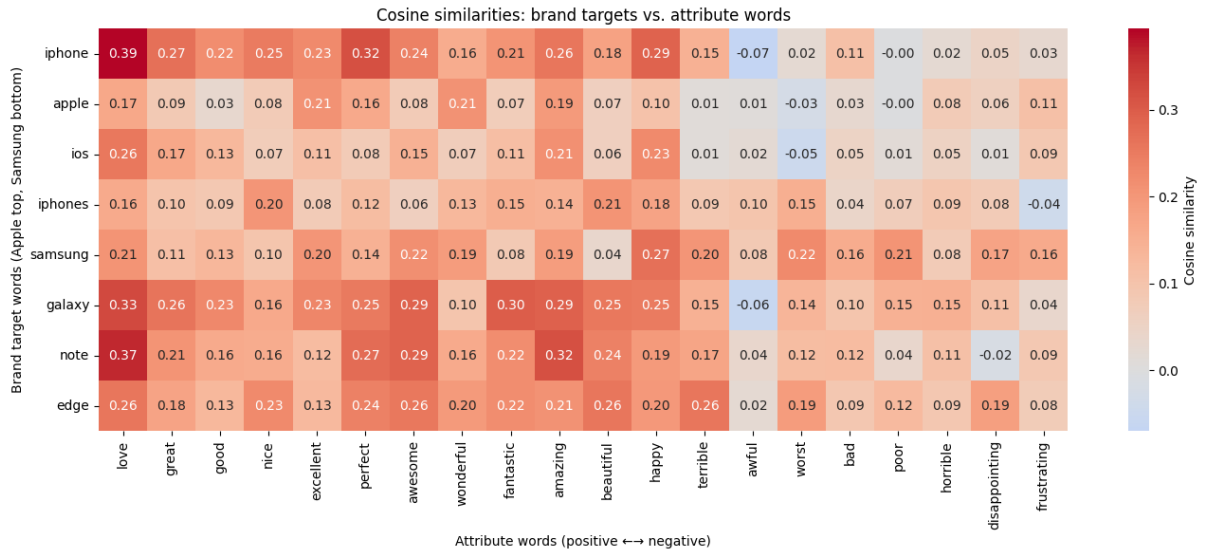


Figure 4.3: Cosine similarities between brand target words and attribute words.

4.3 Discrete-emotion Decomposition

The discrete-emotion analysis revealed that both Apple and Samsung reviews were dominated by positive emotions. This finding is consistent with the generally favorable ratings observed for both brands.

However, differences emerged in the distribution of negative emotions. Samsung reviews contained relatively higher levels of sadness and regret-related language compared to Apple reviews. These emotional patterns suggest that, although Samsung received slightly higher explicit ratings, consumers may have expressed more disappointment or frustration when describing their experiences.

The emotion analysis complements the WEAT findings by providing a more detailed picture of consumer attitudes. Rather than focusing solely on positive and negative evaluations, it highlights specific emotional dimensions that distinguish the two brands.

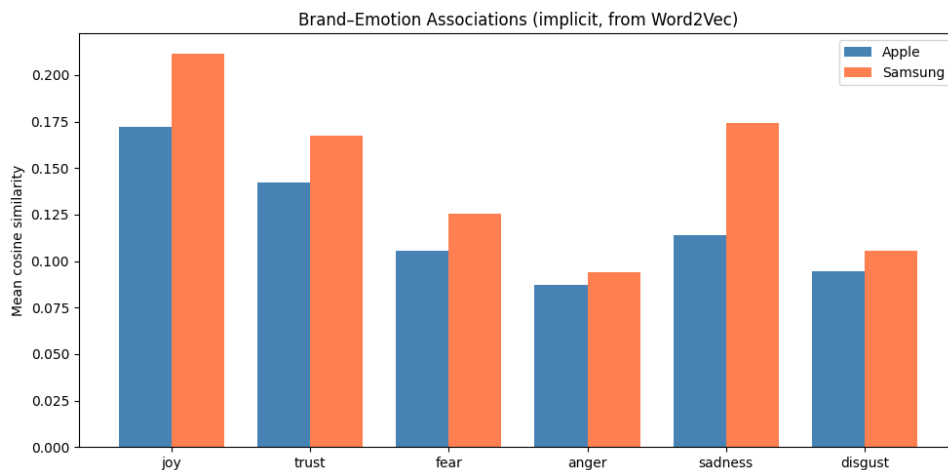


Figure 4.4: Brand-emotion associations derived from Word2Vec embeddings.

CHAPTER 5

Discussion

5.1 Explicit–Implicit Dissociation

The results suggest that explicit and implicit measures do not necessarily capture the same aspect of consumer attitudes. Samsung received slightly higher average ratings, indicating more positive conscious evaluations. However, the WEAT analysis showed stronger positive associations for Apple within the semantic space of the review corpus.

One possible explanation is that consumers use different criteria when assigning ratings and when expressing themselves through language. Ratings may be influenced by practical factors such as product performance, value for money, or satisfaction with a recent purchase. In contrast, implicit associations may reflect broader perceptions connected to a brand, including innovation, prestige, or identity.

Although the WEAT result was not statistically significant, the observed pattern is consistent with the idea that attitudes contain both conscious and automatic components. This highlights the value of examining consumer behaviour from multiple perspectives rather than relying on ratings alone.

5.2 Emotional Differences Between Brands

The emotion analysis showed that both Apple and Samsung were mainly associated with positive emotions, particularly joy and trust. This finding is consistent with the generally favorable ratings observed for both brands.

At the same time, Samsung reviews contained relatively higher levels of sadness and regret-related language. While the difference was not large, it suggests that some consumers expressed disappointment more frequently when discussing Samsung products. These emotional signals may not always be reflected in numerical ratings, demonstrating the additional insight that can be gained from analyzing review text.

The findings indicate that emotion-based approaches can complement traditional measures by revealing aspects of the consumer experience that remain hidden in rating scores.

5.3 Brand-schema Interpretation

The stronger positive associations observed for Apple may be explained by the broader meanings consumers attach to the brand. Beyond product features, Apple is often linked to concepts such as innovation, design, and status. These associations may become embedded in consumer language and contribute to stronger implicit connections within the word embedding space.

The findings suggest that brand perception is influenced not only by product performance but also by the symbolic meanings consumers associate with a brand. As a result, explicit ratings may provide only a partial view of consumer attitudes. Combining behavioural psychology concepts with Natural Language Processing techniques offers a more comprehensive understanding of how brands are perceived in real-world settings.

CHAPTER 6

Limitations

Several limitations should be considered when interpreting the findings of this study.

First, the WEAT analysis produced a moderate effect size but a non-significant p-value. As a result, the observed implicit differences between Apple and Samsung should be interpreted with caution. The relatively small number of target words may have limited the statistical power of the analysis.

Second, the study relied exclusively on Amazon reviews. Consumers who choose to leave online reviews may not be representative of all Apple and Samsung users, introducing the possibility of self-selection bias.

Third, the analyses were based on observational data and therefore cannot establish causal relationships. The results describe patterns in consumer language and ratings but do not explain why these patterns occur.

Finally, the findings are specific to Amazon smartphone reviews. Different platforms or datasets may reveal different patterns of consumer attitudes and brand perceptions.

CHAPTER 7

Conclusion

This study examined explicit and implicit attitudes toward Apple and Samsung using a large collection of Amazon smartphone reviews. Explicit attitudes were measured through customer ratings, while implicit attitudes were investigated using Word2Vec embeddings, WEAT, and emotion analysis.

The results showed that Samsung received slightly higher average ratings, suggesting more positive conscious evaluations. However, the WEAT analysis indicated stronger positive implicit associations for Apple. Although this effect was not statistically significant, it highlights the possibility that explicit evaluations and implicit associations capture different aspects of consumer attitudes.

The emotion analysis provided additional insight by showing differences in the emotional language used when discussing the two brands. Together, these findings suggest that ratings alone may not fully represent how consumers perceive brands.

More broadly, this project demonstrates how behavioural psychology and Natural Language Processing can be combined to study consumer attitudes using large-scale real-world data. Future research could extend this approach by examining additional brands, larger datasets, or alternative sources of consumer-generated content.

Bibliography

- [1] Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- [2] Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.
- [3] Caliskan, A., Bryson, J. J., & Narayanan, A. (2017). Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334), 183–186.
- [4] Amazon Unlocked Mobile Reviews Dataset. Retrieved from: <https://www.kaggle.com/datasets/PromptCloudHQ/amazon-reviews-unlocked-mobile-phones>

APPENDIX A

WEAT Word Lists

This appendix contains the target and attribute word sets used in the Word Embedding Association Test (WEAT).

Target Concepts

- Apple: iphone, apple, ios, iphones
- Samsung: samsung, galaxy, note, edge

Attribute Words

- Positive: love, great, good, nice, excellent, perfect, awesome, wonderful, fantastic, amazing, beautiful, happy
- Negative: terrible, awful, worst, bad, poor, horrible, disappointing, frustrating

The target sets were balanced, containing four representative terms for each brand. Positive and negative attribute words were selected from the review corpus to capture evaluative language commonly used by consumers. Candidate terms that produced unstable associations or appeared primarily in non-evaluative contexts were excluded during exploratory analysis to improve interpretability and reliability.